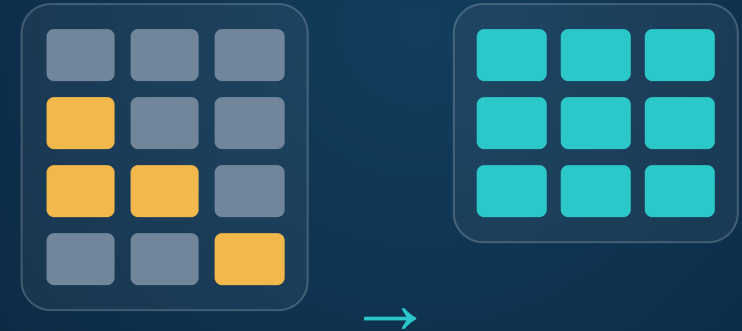


Messy CSV in. Verified dataset out.

A reproducible example showing duplicate removal, explicit selection rules, row reconciliation, and cryptographic evidence.



12

input data rows

4

duplicates removed

8

validated output rows

One rule, fully reconciled

KEY: customer_id · KEEP: last

INPUT SAMPLE

Duplicate customer records

ID	NAME	PLAN	UPDATED
C-1001	Alice Carter	basic	Jun 01
C-1002	Ben Turner	pro	Jun 02
C-1005	Emma Wilson	basic	Jun 05
C-1008	Henry Adams	basic	Jun 08
C-1002	Ben Turner	business	Jun 10
C-1005	Emma Wilson	pro	Jun 11
C-1001	Alice Carter	pro	Jun 12
C-1008	Henry Adams	business	Jun 13

CLEAN OUTPUT

Latest record retained deterministically

ID	NAME	PLAN	UPDATED
C-1003	Chloe Martin	basic	Jun 03
C-1004	Daniel Wright	pro	Jun 04
C-1006	Finn Parker	basic	Jun 06
C-1007	Grace Collins	pro	Jun 07
C-1002	Ben Turner	business	Jun 10
C-1005	Emma Wilson	pro	Jun 11
C-1001	Alice Carter	pro	Jun 12
C-1008	Henry Adams	business	Jun 13

12

rows read

8

rows written

4

duplicates removed

0

unexplained row changes

Reproducible by design

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1 Validate structure

Reject blank or duplicate headers, missing key columns, malformed rows, and ambiguous output paths.

2 Apply the agreed rule

Deduplicate by `customer\_id` and retain the last source occurrence. Source order remains deterministic.

3 Reconcile and hash

Record input, output, removed-row counts, parameters, and SHA-256 hashes in a JSON audit summary.

- ✓ Original input remains untouched
- ✓ Explicit keys and retention rule
- ✓ Machine-readable audit summary

```
$ csv-cleaner contacts-before.csv contacts-clean.csv \  
  --keys customer_id --keep last \  
  --summary audit-summary.json
```

```
rows_read:      12  
rows_written:   8  
duplicates_removed: 4  
key_columns:    customer_id  
keep:           last
```

SHA-256 evidence

Input

8df280cba47dbd2be78171ef03bfcf37421b0f9d04ed278d9dc266094c4f8bce

Output

7f56fe5753e40232caa4c9bd68612252f60d6d42188404b51464c4051f4447e4

- ✓ Normalized UTF-8 output
- ✓ No network calls or telemetry
- ✓ Eight automated unit tests